

**UNDERGRADUATE RESEARCH
OPPORTUNITIES PROGRAM (UROP) PROJECT
REPORT**

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Summer Semester, 2024

Abstract

To validate the existing theoretical and empirical studies, a computation model of the annular gas-liquid flow regime was developed. This study employed the Volume of Fluid (VOF) method by using the ANSYS Fluent to simulate the gas-liquid flow in a 50 mm radius pipe with air and water at the standard temperature and pressure (STP). The model is coupled with SST turbulence and 5 m length to observe the development of the flow. By utilizing the data from the NUS Multiphase Flow Loop Lab and the flow regime map proposed by Taitel and Dukler (1976) as the boundary conditions. According to the simulation, a wavy annular flow regime is found

Preface and acknowledgement

I would like to extend my gratitude to Professor W. L. Loh for his invaluable guidance throughout this project and in the development of this work. I also wish to acknowledge Ph.D. L. Yu of the National University of Singapore for his inspiring advice in the fields of computational modelling and fluid mechanics.

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1.0 Introduction

In recent years, the field of fluid mechanics has thoroughly characterized various flow regimes through extensive experimental and theoretical modelling. These models and data help engineers make better predictions of the multiphase flow under different conditions. Moreover, computational fluid dynamics (CFD) can be also used to predict the liquid and gas flow based on the physics equations. It helps provide more accurate numerical modelling to assist in engineering design.

The study mainly focuses on using the Volume of Fluid (VOF) method, which is an accurate way of tracking two or more fluids in each cell throughout the computational domain. In this report, the VOF method is used to simulate annular gas-liquid flow in a horizontal pipe. Here, “horizontal” signifies that the flow is oriented perpendicular to gravitational forces. Annular flow is widely used in the industry due to its ability to substantially reduce pressure losses, leading to more efficient oil transportation.

This research will prioritize verifying existing theories and experimental data of liquid-gas regimes from the NUS Multiphase Flow Loop Lab by using a CFD model. This simulation process will help gain a more comprehensive understanding of annular flow regimes and enhance the accuracy of the liquid-gas flow model. It also helps the engineers extend to applications beyond the oil and gas sector such as liquid-liquid regimes.

Additionally, the research includes an investigation of mesh independence to ensure the accuracy and reliability of the CFD simulations. Mesh independence study is a crucial method of investigating the simulation result. It confirms that the result will not be influenced by the discretization of the computation domain. By thoroughly analyzing mesh independence, this study aims to reinforce the precision of the VOF method applied to annular multiphase flow simulation.

2.0 Theoretical Perspective

2.1 VOF Method

The VOF method employs a fixed Eulerian mesh to track surfaces by solving a single set of momentum equations coupled with a volume fraction equation. Each mesh should have a fractional value between zero and one to determine the phase fractions in every computational cell, where one represents one kind of fluid, and zero represents another. By using the volume fraction equation, we can define the different phase properties in different regions.

$$\frac{\partial u_i}{\partial t} + \nabla(cu) = 0$$

Moreover, by applying the VOF method with the SST $k - \omega$ turbulence, the project achieved a more reliable and detailed simulation of the gas-liquid annular flow. The VOF method can effectively capture the interface dynamics, while the SST $k - \omega$ turbulence model ensures an accurate representation throughout the flow domain. The synergy between the two models allowed for the prediction of flow patterns and closely matched the experimental observation from real life. The two models demonstrate the effectiveness of the combined approach in studying the multiphase flows.

2.2 Superficial Velocity and Multiphase Flow Regime Map

The superficial velocity is the velocity of fluid moving through a pipe, it is defined as the volumetric flow rate of that fluid divided by the cross-sectional area. The equation is shown below:

$$J = \frac{Q}{A_c}$$

With the superficial velocity, we can define the regimes of the multiphase flow by using the map proposed by Taitel and Dukler in the mid-1970s. Flow regime maps define the various flow

regime transitions based on experimental data or mechanistic models (Figure #). Typically, different pressure drops correlations start with distinct flow maps, as the relationship between pressure drop and flow regime is established during the experimental phase of the research. As a result, Once the respective superficial velocities and specific environment are determined, the flow regime can be readily identified from the graph.

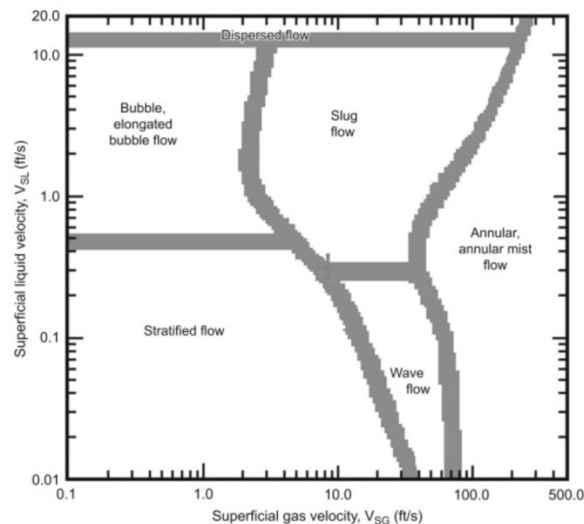


Figure #. The Flow Regime Map

2.3 Mesh Independence

The grid independence test is a process used to determine the optimal grid configurations that use the fewest number of grids while ensuring that the numerical results are consistent across various grid conditions. As the example (Figure #) shows, when the number of the mesh is larger than a certain value, the result does not change with the quantity of the mesh. In addition, the mesh independence study process involves systematically refining the mesh and comparing the results to ensure that further refinement does not significantly alter the outcomes (Figure#).

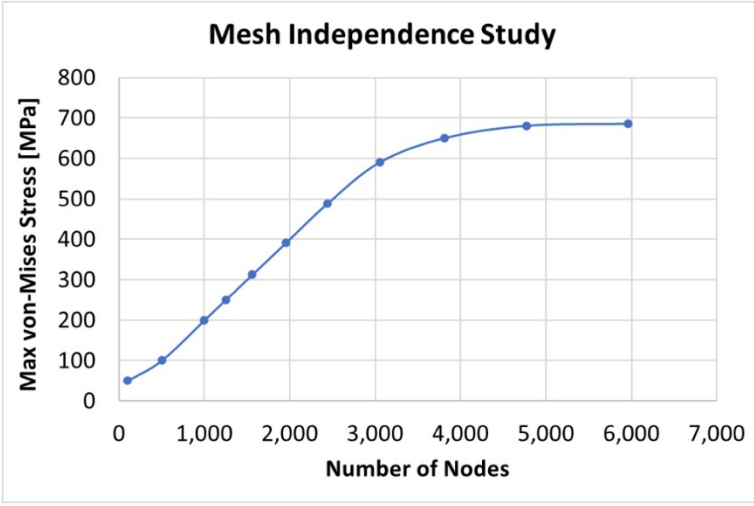
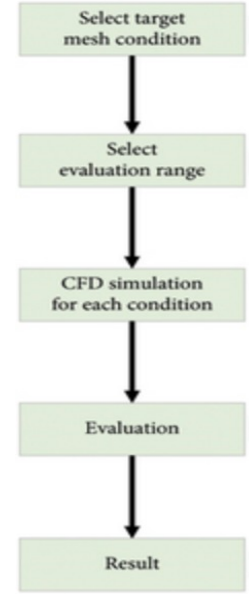


Figure #. The Example of Mesh Independent Study



Figure#. The Process of Study

3.0 Methodology

While a more precise model might be developed by using an Eulerian multiphase approach, the associated computational expense and complexity of model construction would be substantially greater. As a result, the VOF model was constructed by first analyzing high-speed footage of the flow regime captured at the NUS Multiphase Flow Loop Lab (Figure #) and incorporating superficial velocities measured during the experiment

The geometry of the pipe (Figure #) used in the experiment is designed with an outer diameter of 50mm and a 40 mm inlet diameter throughout the study. The length of pipe is constructed with a length of 5 m to ensure the observation of a more fully developed flow regime

The superficial velocities reported in the experiment were $J_a = 25 \frac{m}{s}$ and $J_w = 0.05 \frac{m}{s}$. By using equation# and the data of the geometry, the average velocity normal to the inlet for each phase was determined. The yielded inlet velocities are then $V_a = 39.06 \frac{m}{s}$ and $V_w = 0.139 \frac{m}{s}$.

According to the superficial velocities and the flow regime map (Figure #), we can find that inlet conditions from the experimentation produce an annular flow. While the model resulted in a

dispersed water-in-air flow regime when applied. To get a reasonable result, we increase the superficial velocity of the water phase to $J_w = 0.45 \frac{m}{s}$.

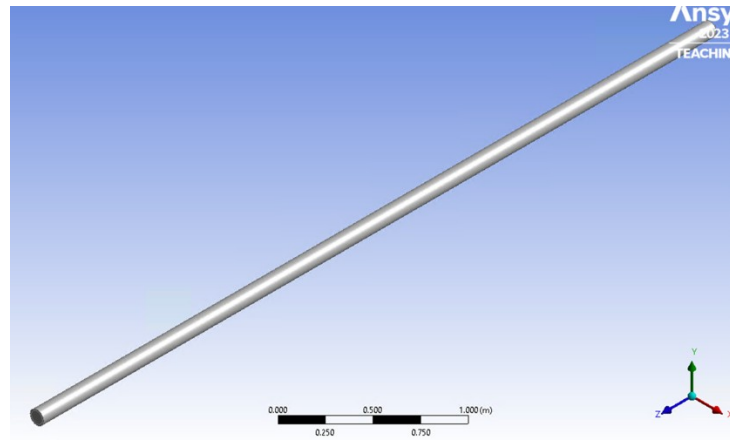


Figure #. The Geometry of the Pipe

4.0 Presentation of Findings and Data

The full-scale study results reveal a wavy annular flow regime with the water droplets within a short pipe segment between 1 and 2 meters. By comparing the high-speed footage from the NUS lab and the simulation result, several key observations can be made.

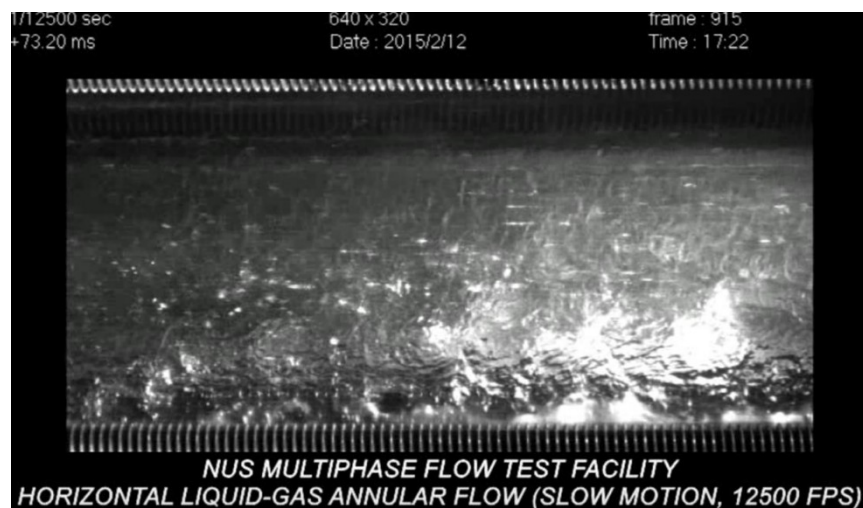


Figure #. Annular Flow observed in the NUS Multiphase Flow Test Facility

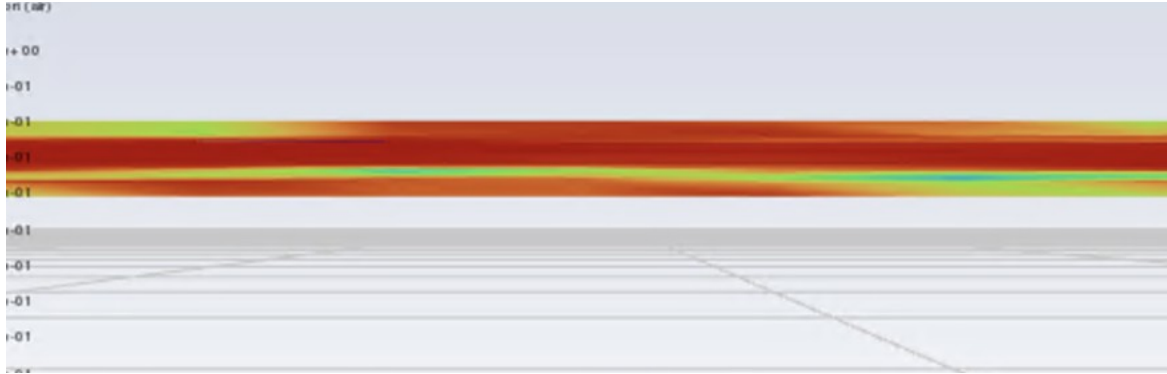


Figure #. The phase contour simulation results about the midplane between 2.5 m to 3m

The contour plot employs color coding to illustrate the volume fraction of air and water, with red indicating pure air and green and yellow signifying pure water. The computational model reveals the experimental observations in different aspects. First, the water flow is biased towards the bottom of the pipe. Although in the beginning, the difference in the amount of water above and below is not large (Figure#), as the length of the pipe becomes longer, this phenomenon begins to become obvious. Moreover, the air-water interface in both the computational model and the experimental observations exhibits wavy characteristics with distinct ripples. However, the presence of these ripples varies in intensity and frequency, potentially due to differences in mesh resolution and the turbulence model parameters used in the simulations.

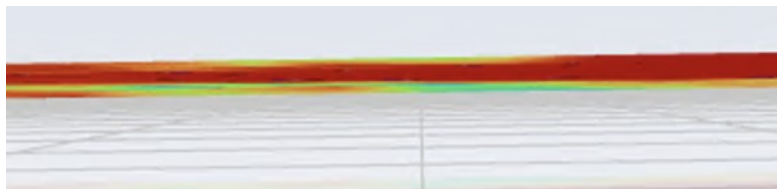


Figure #. The phase contour simulation results about the midplane between 0.25 m to 0.5m

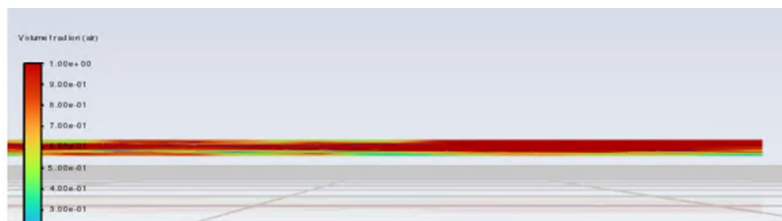


Figure #. The simulation result from the 3 m to 5m

From Figure #, the wavy flow can be seen more clearly along the relatively long distance.

5.0 Data Analysis and Discussion

The first phenomenon which water flow is biased towards the bottom of the pipe can be explained by the principles of buoyancy effects which is a tendency of an object to float or to rise in a fluid when submerged. Buoyancy forces arise due to the density difference between the two fluids -air and water in this case. Water, being denser than air, naturally settles at the bottom of the pipe under the influence of gravity.

For the second phenomenon, the air-water interface is an annular flow regime that exhibits wavy characteristics with ripples, which is due to the complex interactions between the gas and liquid phases. In an annular flow, the liquid phase forms a continuous film along the pipe wall, while the gas phase flows through the core of the pipe. The interface between these two phases is not smooth but instead displays a wavy and ripple-like feature due to various hydrodynamic factors.

However, from the figure # and figure #, there is still a difference between the results. In the simulation results, the ripples are barely seen. These discrepancies are likely attributable to various factors related to the model setup and mesh resolution. The smoother interface in the model is probably a consequence of a coarse mesh and insufficiently refined parameters in the turbulence model. The unclear wavy flow may result from both the mesh size and the increased superficial velocity. Additionally, distinguishing between entrained droplets and interface distortions presents a challenge.

6.0 Conclusion

The conclusion of this research report underscores the efficacy of using the Volume of Fluid (VOF) method in conjunction with the SST $k-\omega$ turbulence model to simulate annular gas-liquid flow in a horizontal pipe. The computational model developed closely mirrors experimental observations, demonstrating a wavy annular flow regime with entrained droplets, with some discrepancies likely due to mesh resolution and turbulence model parameters. The study highlights the importance of mesh independence in ensuring accurate and reliable CFD simulations. Furthermore, the findings validate existing theories and experimental data, offering valuable insights into the complex dynamics of annular flow regimes and their practical applications in industries beyond oil and gas, such as liquid-liquid systems. This research contributes to a deeper understanding of multiphase flow behaviours and supports the development of more precise and efficient flow engineering designs.